

ML Week 10

Number of participants: 24



1. Neural Networks

7 correct answers
out of 19 respondents



A neural network where all activation functions are the identity function is a linear model.



11 votes

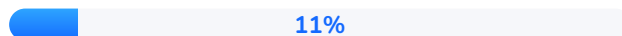


The flexibility of neural networks comes from the non-linearity of the activation functions.



15 votes

Neural networks have no tuning parameters.



2 votes

A neural network of fixed width and depth can approximate any function arbitrarily well.



5 votes



2. Which of the following statement(s) is/are true about neural networks?

14 correct answers
out of 20 respondents



Can be used for classification problems with arbitrarily many classes.



19 votes

Always converge to the global optimum, if trained for enough epochs.



2 votes



Can approximate any continuous function on the predictor space arbitrarily well if the number of nodes is large enough.



15 votes

Given a regression problem with

$p = 40$ predictors, what is the



3. number of parameters for a neural network with $M = 2$ hidden layers containing 60, and 30 neurons respectively?

10 correct answers
out of 14 respondents

4230



2 votes



4321



10 votes

72000



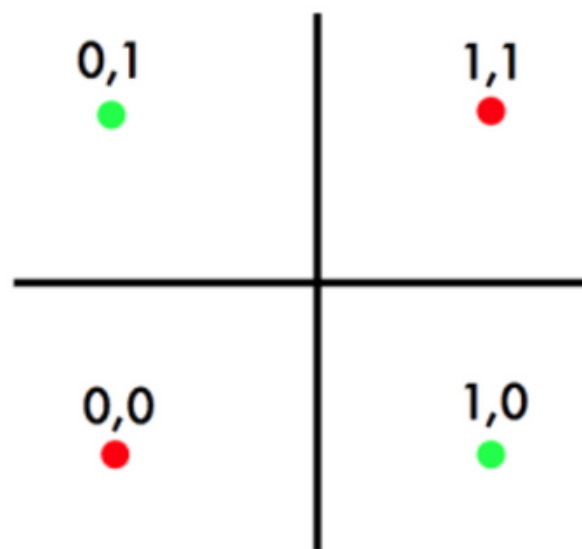
2 votes

Consider the dataset in the picture, where red and green points correspond to positive and negative classes, respectively. Let $g(x) =$



4. $\text{ReLU}(x) = x1\{x > 0\}$ be the activation function. What is the smallest number of layers M to perfectly classify the dataset (including the input layer)?

11 correct answers
out of 16 respondents



$M = 1$



1 vote



$M = 2$



11 votes

$M = 3$



4 votes